



Amazon SDE Interview Experience

About the company: Amazon, one of the **MAANG** companies, is a global leader in e-commerce, cloud computing, and AI. Known for its innovation and customer obsession, Amazon empowers engineers to build scalable, high-impact solutions that serve millions of users worldwide.

Role Offered: Software Development Engineer Intern

Stipend: 1,10,000 INR + 425 USD (Relocation Allowance) per month

Duration: 6 Months

Location: Bengaluru, Hyderabad, Chennai (Available Locations)

Selection Process:

1. Online Assessment
2. Technical Interview (Virtual)

Online Assessment – 3rd October 2025

The Online Assessment was conducted on HackerRank platform with a duration of 70 minutes (coding test). The Online Assessment had 3 sections:

1. Coding Assessment (2 questions)
2. Workstyle Assessment
3. Work Simulation Assessment

1.Coding Assessment (70 Minutes):

Coding Question 1:

Problem Statement:

You are given an array representing items stored in different warehouses.
In one operation, you can:

- Dispatch two items if they belong to different warehouses, or
- Dispatch one item if it's the only one remaining.

Find the minimum number of operations required to dispatch all items.

Input: [2, 9, 7, 8, 8]

Output: 3

Explanation:

- Pair (2, 8)
- Pair (9, 7)
- One item (8) left alone

Total operations = 3

Approach:

1. Count frequencies of each warehouse ID using a hashmap.
2. Find the maximum frequency (maxFreq) among all warehouses.
3. Let n be the total number of items.
 - o If $\text{maxFreq} \leq n / 2$: all can be paired efficiently $\rightarrow$ operations = $\text{ceil}(n / 2)$.
 - o Else: some items remain unpaired $\rightarrow$ operations = maxFreq.
4. Return the maximum of these two values.

Coding Question 2:

Problem Statement:

You are part of Amazon's Network Infrastructure Division and are working on a project to maximize data transfer efficiency across its network of data centers. You are given n data centers, and the bandwidth capacity of each center is given in an integer array named capacity.

There are connection count data streams that need to be routed through two data centers — one as the primary route and the other as the failover route. Each data stream must use a unique pair of data centers for its routing.

The bandwidth rate for each data stream is defined as the sum of the capacities of its primary and failover data centers.

You need to find the maximum total bandwidth rate that can be obtained by optimally choosing unique pairs of routes for each data stream.

Note:

- A pair (x, y) is unique if no other stream has selected the same pairs.
- Pairs (x, y) and (y, x) are treated as different routes.
- Selecting the same data center for both routes (x, x) is also valid.

Input:

capacity = [5, 3, 6]

connectionCount = 4 (Total Number of pairs to be taken)

Output: 44

Explanation:

All possible pairs of data centers are:

[1,1], [1,2], [1,3], [2,1], [2,2], [2,3], [3,1], [3,2], [3,3]

To achieve the maximum total bandwidth Rate, we can select the pairs:

[3,3], [3,1], [1,3], [1,1]

The total bandwidth rate = $(6+6) + (6+5) + (5+6) + (5+5) = 44$

Approach:

1. Sort the capacity array in descending order, because the highest bandwidth sums will come from the largest capacities.
2. Use a max heap to efficiently fetch the largest pair sums without generating all possible pairs (which would take $O(n^2)$).
3. Start by inserting the pair (0, 0) (the largest element with itself) into the heap.
4. Each time we pop the largest sum from the heap:
5. Add its sum to the total bandwidth.
6. Push the next two possible pairs: (i + 1, j) and (i, j + 1) into the heap (if not already used).
7. Keep repeating until we have selected connectionCount pairs.
8. The total sum collected after these pops is the maximum total bandwidth rate.

Similar Problem: <https://leetcode.com/problems/find-k-pairs-with-smallest-sums/description/>

2. Workstyle Assessment:

A short personality and behavioural test based on **Amazon's Leadership Principles** — such as *Customer Obsession*, *Ownership*, *Bias for Action*, *Deliver Results*, etc.

You'll be shown statements like:

- “I prefer to take initiative even when tasks are not assigned to me.”
- “I often find ways to simplify complex problems.”

You must choose which statement describes you more accurately.

Tips:

- Be consistent with answers (don't contradict yourself).
- Show ownership, teamwork, customer focus, and curiosity.
- Avoid extremes like “always disagree” for everything.
- Read **Amazon's Leadership Principles** before the test — it helps you understand what traits they value.

3. Work Simulation Assessment:

A situational judgment test where you're placed in real-world scenarios faced by SDEs or team members at Amazon.

Example:

“You're working on a project with a tight deadline, and a teammate reports a critical bug. What would you do first?”

You'll select how you would respond or prioritize tasks from multiple options.

Tips:

- Prioritize customer impact and data-driven decisions.
- Show ownership (take responsibility instead of blaming).
- Balance speed and quality (“Deliver Results” and “Insist on Highest Standards”).

Both tests are screening filters used before or after the coding assessment, depending on the hiring flow.

- Even if your coding test is good, poor performance here can stop your shortlisting.

- Amazon looks for technical + cultural fit.

So, it's very important to take both seriously and align your responses with **Amazon's Leadership Principles**.

As for selection, there isn't a fixed factor that guarantees shortlisting. Luck also plays a major role.

Based on this Online Assessment, 107 people were shortlisted for the interview.

Technical Interview (1 Hour): - (10th October 2025)

The interview was conducted through Amazon Chime, where the interviewer shared a Live Coding link a shared text editor where both of us could write and edit code (execution was not supported). The interviewer, who also happened to be an alumnus of CEG, started the session by asking me to introduce myself, talk about my projects, and share a bit about my college life.

The interaction felt friendly and engaging. After the discussion, the interviewer pasted a coding problem in the editor, and I had to explain my approach and write the optimized solution.

Question 1: Product Discount Price Calculation:

Problem Statement:

You are given an array $a[i]$ where each $a[i]$ represents the price of a product. For each product, you need to find the first element to the right that is smaller than or equal to the current product price.

That element represents the discount for the current product.

If such an element exists, the final price for that product = $a[i] - \text{discount}$.

If no such element exists, the product price remains the same ($a[i]$).

Finally, return the sum of all final prices.

Input: [8, 4, 6, 2, 3]

Output: 15

Explanation:

- For 8, next smaller/equal = 4 $\rightarrow$ discount = 4, final price = $8 - 4 = 4$

- For 4, next smaller/equal = 2 $\rightarrow$ final price = $4 - 2 = 2$
- For 6, next smaller/equal = 2 $\rightarrow$ final price = $6 - 2 = 4$
- For 2, no smaller/equal $\rightarrow$ final price = 2
- For 3, no smaller/equal $\rightarrow$ final price = 3

Total sum = $4 + 2 + 4 + 2 + 3 = 15$

Approach (Monotonic Increasing Stack - Optimized $O(n)$)

The interviewer first asked me to explain my approach before writing the code.

I explained my approach to the interviewer like this:

- I used a monotonic increasing stack to efficiently find the next smaller or equal element to the right for each price.
- I iterated through the array from left to right.
- For each element:
 - While the stack is not empty and the top element is greater than or equal to the current price, it means we found a discount for that previous element — so we pop it and calculate the discounted price.
 - Finally, I pushed the current price into the stack.

After the iteration, if any elements remain in the stack (i.e., no discounts found for them), add all remaining stack values to the final value.

Question 2: Minimum Moves in a Grid with Obstacles:

Problem Statement:

You are given an $n \times n$ matrix containing only 0s and 1s, where:

- 0 $\rightarrow$ water (you can move through it)
- 1 $\rightarrow$ obstacle (you cannot pass through it)

You are also given a value k .

Starting from the top-left corner (0, 0), you need to reach the bottom-right corner ($n-1$, $n-1$) using the minimum number of moves.

In one move, from your current cell (i, j) , you can move up, down, left, or right — but up to k steps in that direction, stopping only before hitting an obstacle (1) or the boundary.

You need to return the minimum number of moves required to reach the destination.

If it's not possible, return -1.

Approach (BFS)

I explained my approach to the interviewer like this:

- Since we need the minimum number of moves, BFS is the best choice.
- Each cell acts like a node in a graph, and possible jumps (up to k steps) are the edges.
- From every current cell (r, c) :
 1. Try moving up, down, left, right — up to k steps.
 2. Stop exploring further in that direction if:
 - You hit a boundary,
 - You hit an obstacle (1), or
 - You reach a cell that's already visited with fewer moves.
 3. Each valid new cell is pushed into the BFS queue with moves + 1.
- Continue BFS until reaching $(n-1, n-1)$ or the queue becomes empty.
- Return the number of moves if reachable, else -1.

After I completed both coding questions, the interviewer told me that he was satisfied with all my approaches and appreciated the clarity in my explanation and thought process.

He then asked if I had any questions for him.

I asked him about the work culture at Amazon and how his life was after graduating from CEG.

He responded very kindly and shared his experiences, explaining how Amazon encourages ownership, collaboration, and constant learning. It was a very pleasant and insightful conversation.

Overall, it was one of my best experiences, and I genuinely enjoyed the interaction — especially because the interviewer was also an alumnus of (CEG), which made the discussion feel even more comfortable and relatable.

This interview was a truly memorable experience for me — from the **Online Assessment** to the **final discussion** with the interviewer.

For anyone preparing for placements, my honest advice is:

1. **Practice regularly** — consistency matters more than long hours.
2. **Solve problems daily on LeetCode and GeeksforGeeks** to strengthen your understanding of patterns.
3. **Participate in coding contests** — they help you think fast under pressure and identify your weak spots.

If things are not going the way you expected, **trust the process** — it might be happening in a better way than you can imagine.

Never feel low at any stage. Every rejection or tough question is just a **step closer to success**.

“God’s plan is always the best plan ♥”

Start your preparation early, stay consistent, and believe in yourself — your time will come!!!

All the Very Best!!!

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